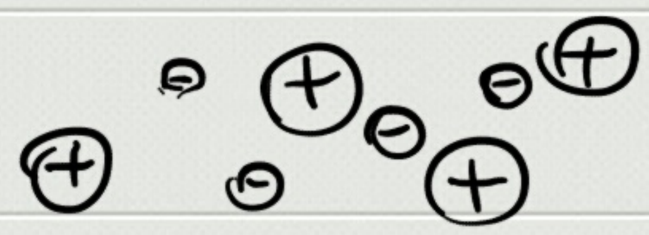
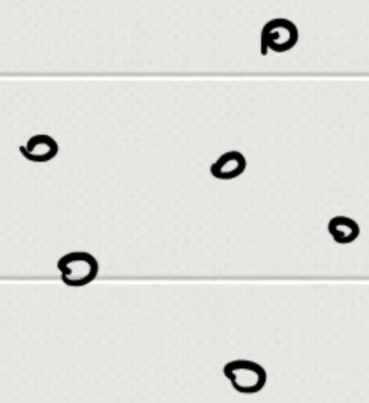
Plasma 

Solido forma + volume

Liquido volume

Gas  —

O $\varnothing$ 1 cm $\Rightarrow$ 27 in 1 m^3

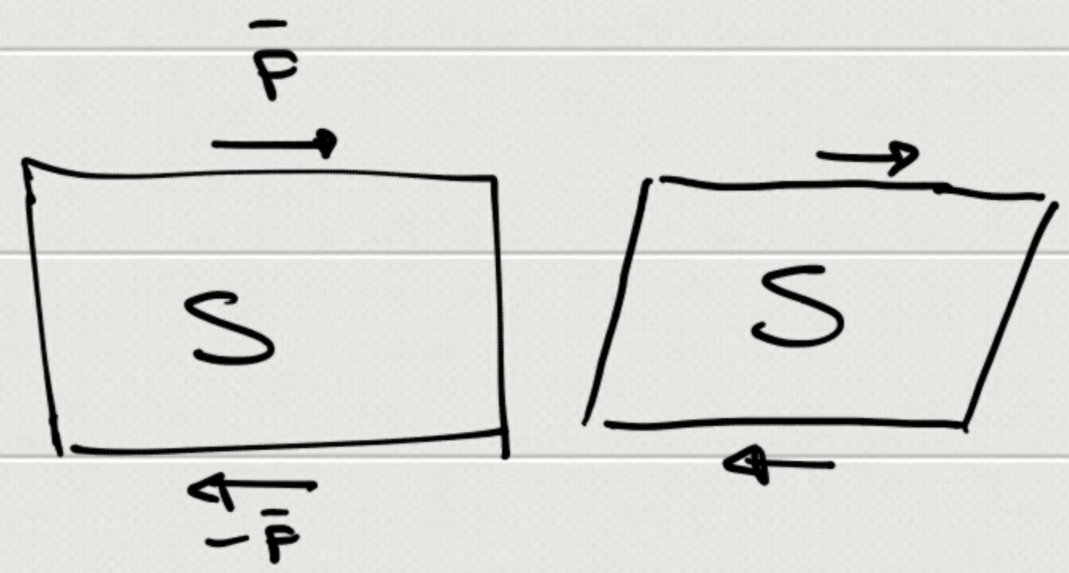
Fluido : liquido / gas

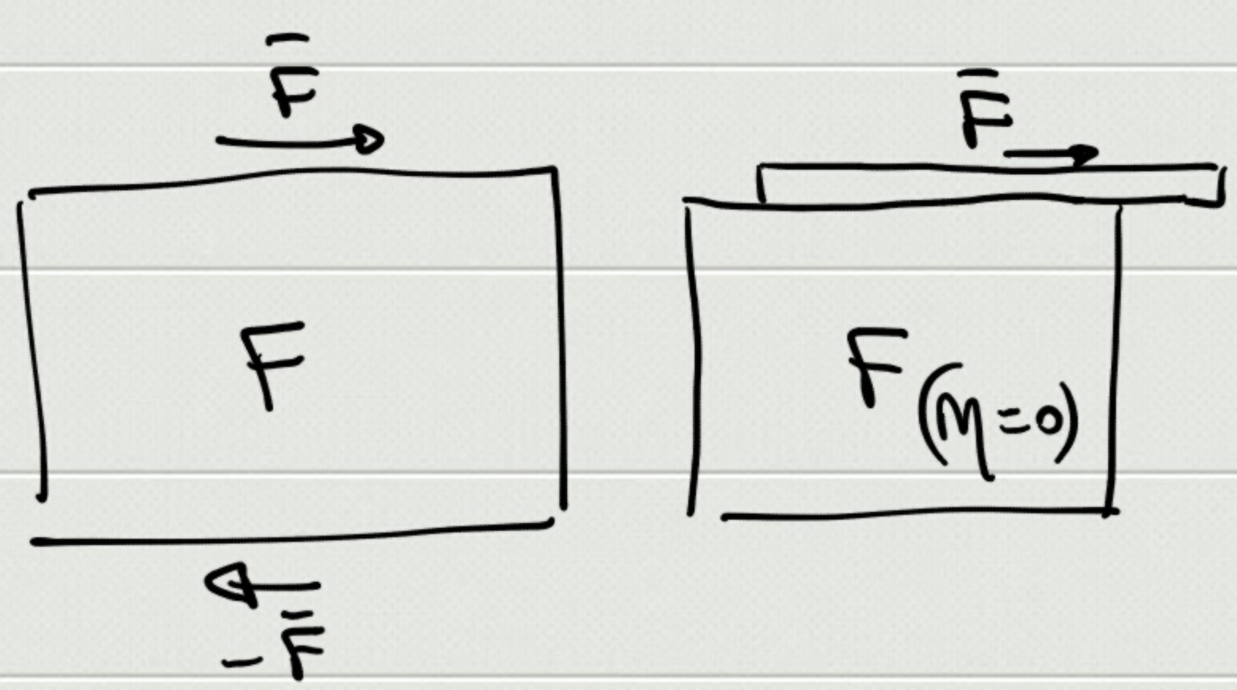
$\sim$ dV, dm (continuo)

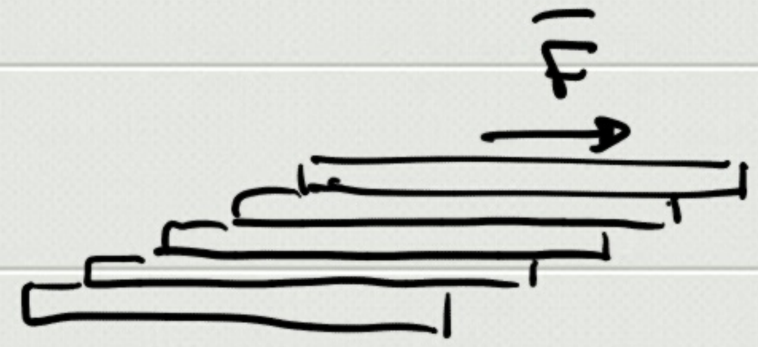
fluido ideale $\boxed{\rho = cost}$ viscosità nulla

$\boxed{\eta = 0}$

No sforzi di taglio



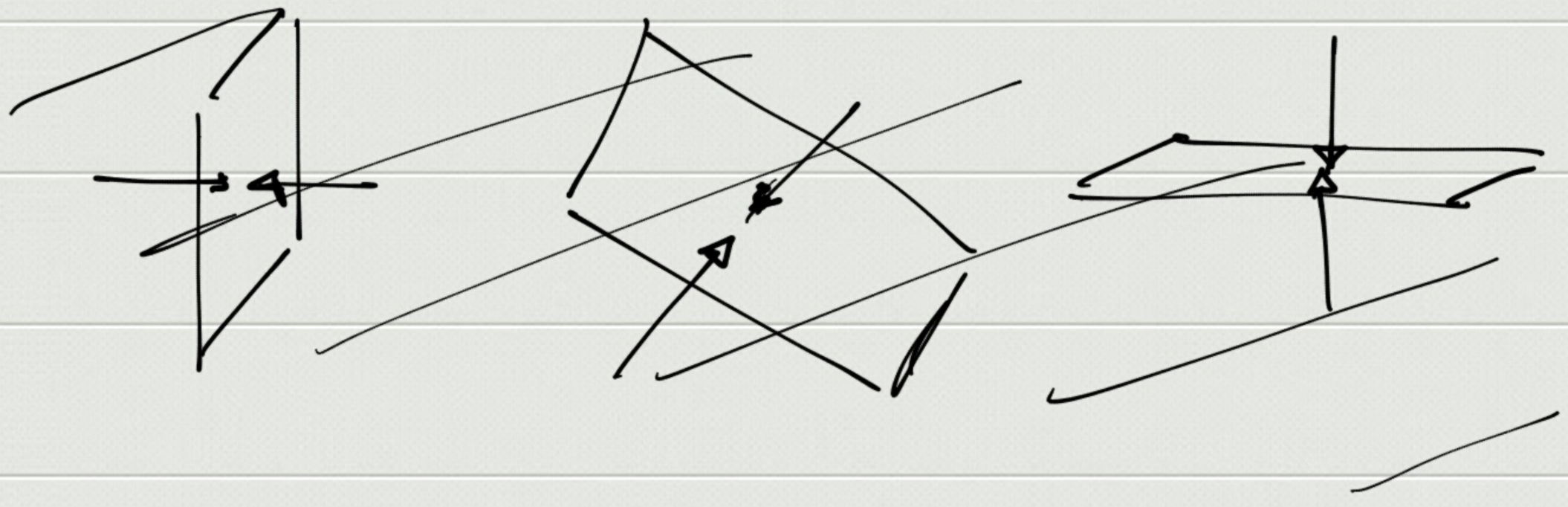


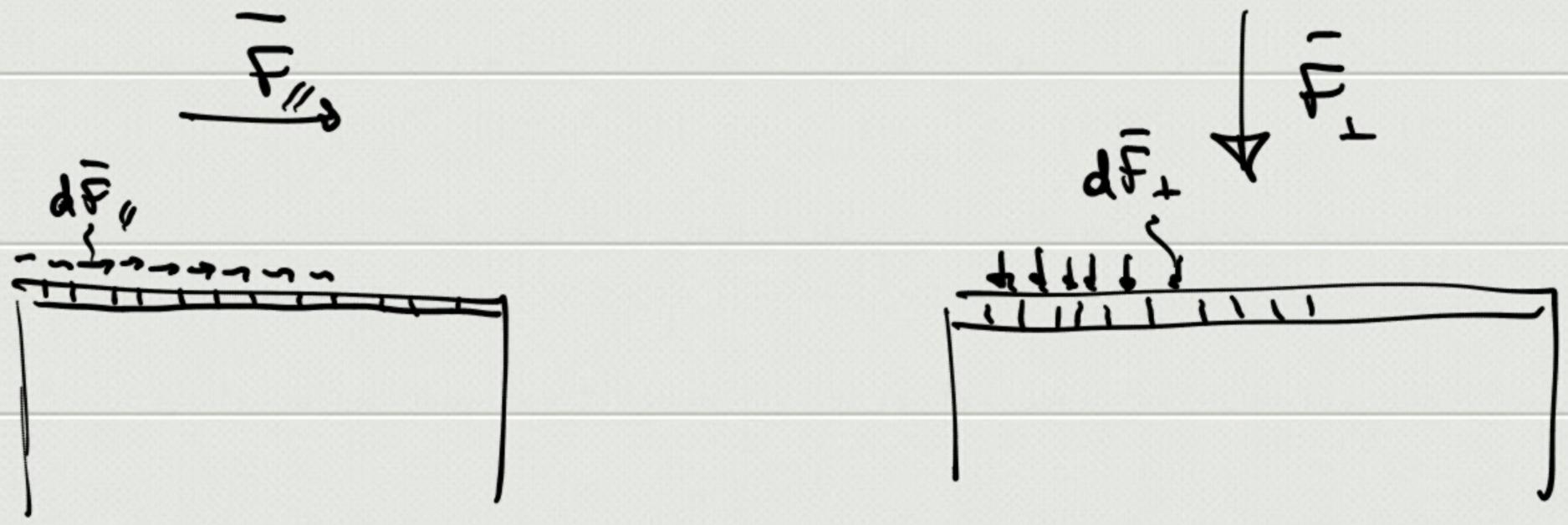


$\eta \neq 0$

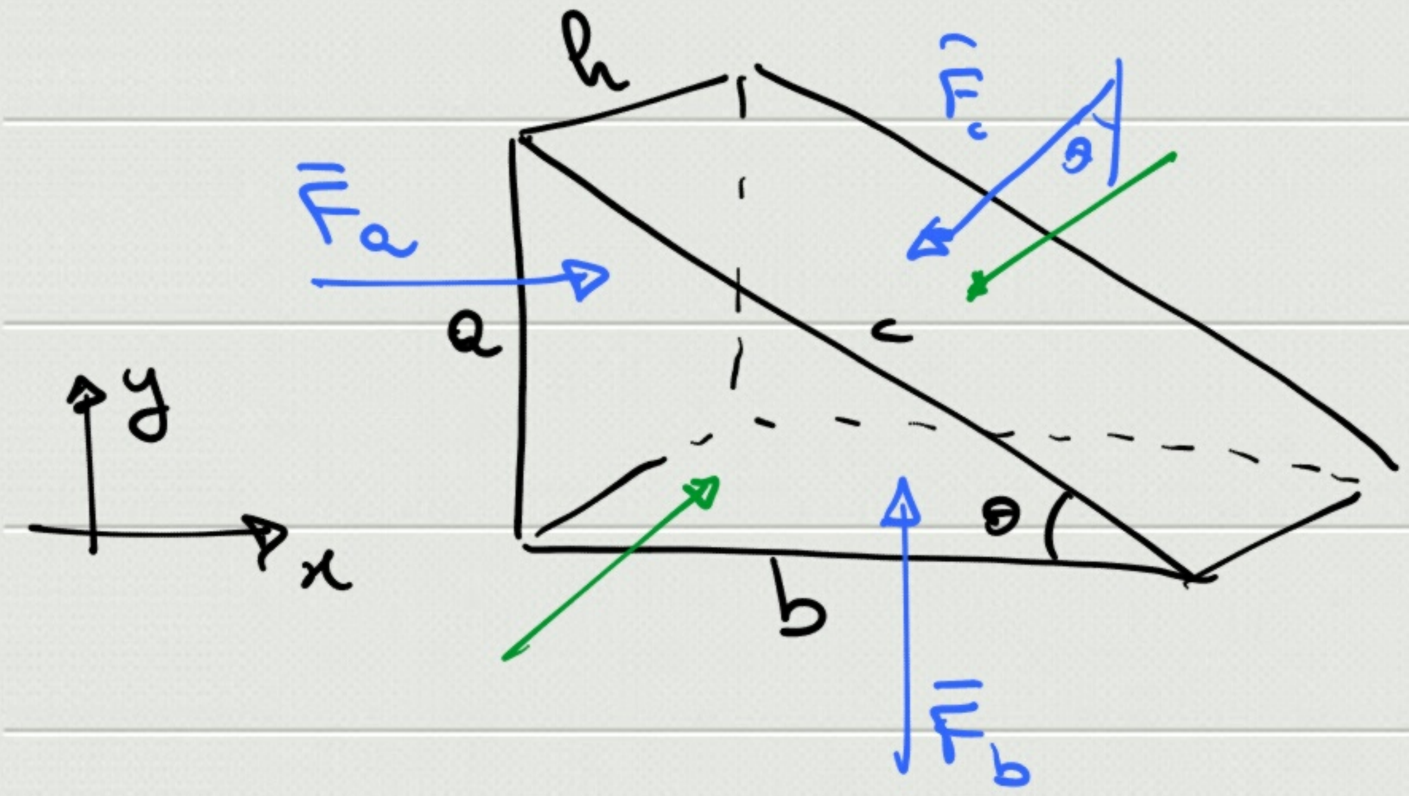
Fluido <u>statico</u> $\Rightarrow$ NO sforzi di taglio

$\Rightarrow$ Solo forze perpendicolari alla superficie





pressione : $\boxed{p = \frac{F_{\perp}}{S}}$ (scalare)



Statica

$$\bar{F}_a + \bar{F}_b + \bar{F}_c = 0$$

$x: \quad F_a - F_c \sin\theta = 0 \quad \Rightarrow \quad p_a \cancel{a}\cancel{h} = p_c \cancel{c}\cancel{h} \cancel{\sin\theta}$

$y: \quad F_b - F_c \cos\theta = 0 \quad \Rightarrow \quad p_b \cancel{b}\cancel{h} = p_c \cancel{c}\cancel{h} \cancel{\cos\theta}$

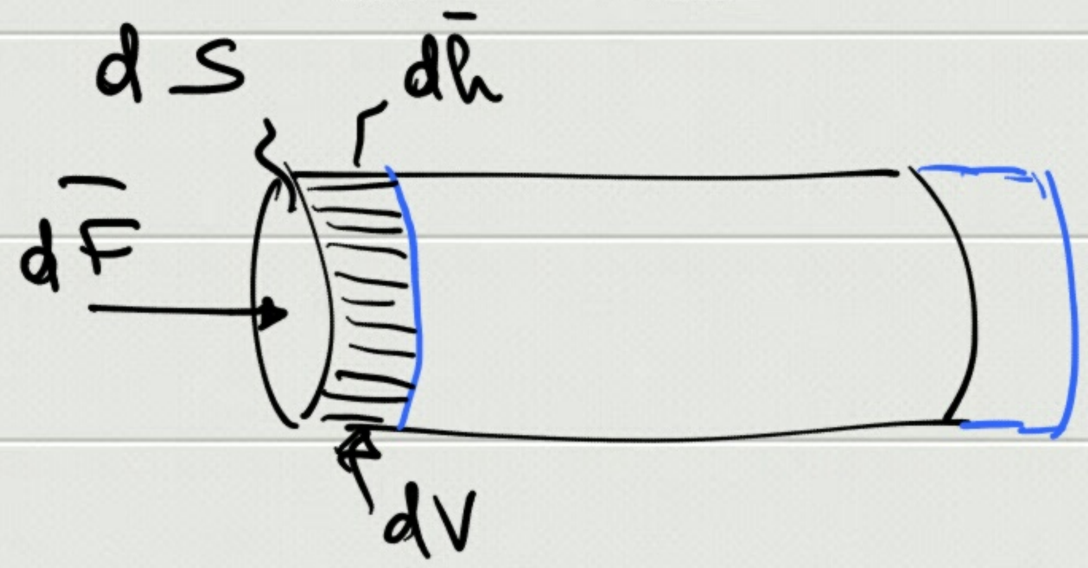
$$\Rightarrow p_a = p_c \qquad p_b = p_c \quad \Rightarrow \quad \boxed{p_a = p_b = p_c}$$

$$\boxed{p = \frac{dF_\perp}{dS}}$$

$$[p] = \left[\frac{F}{S}\right] = \frac{N}{m^2} = Pa$$

(Pascal)

$$1\ atm = 1.01325 \cdot 10^5\ Pa$$



$$dW = d\bar{F}\, d\bar{h} = p\, dS\, dh = p\, dV$$

$$\boxed{W_{i \to f} = \int_i^f p\, dV}$$